ALLIANCE UNIVERSITY | NAAC GRADE A+ ACCREDITED UNIVERSITY

MINI PROJECT | PROJECTION + HSR

3D Airport Terminal: Mission Gate B1

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OBJECTIVE & OUTCOME

A walkable 3D airport terminal built in Blender with one Python script, using perspective projection and hidden surface removal for real-time navigation and a playable mission.

80 × 30 m
terminal floor, 12 m ceiling, glass facades front and back

18
structural columns plus glass walls

16
animated NPCs that walk, idle and talk

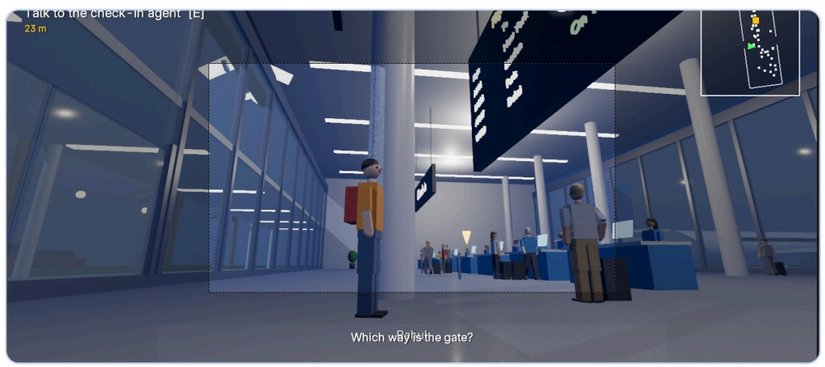
3
start modes: PLAY, TOUR and WALK

1 Step off the curb: the game



Airport forecourt with live objective and minimap HUD

2 Inside the check-in hall



Departures board, check-in desks and talking NPCs

3 The mission

1. **Talk to the check-in agent**
   Walk to the desks and press E. A distance counter shows how far you are.
2. **Go through Security**
   Follow the minimap yellow marker through the terminal.
3. **Talk to the officer**
   Subtitles and voice lines react to the player.
4. **Reach Gate B1**
   Clear the last objective and make your flight.

4 Airport gallery from the game

Terminal facade



Check-in desks



Departures board



Player avatar



Captured in Blender: glass walls, columns, animated people

5 Projection + hidden surfaces

Model space › World space › Camera space › Screen space

$$x_s = W/2 + f \cdot x / z$$
$$y_s = H/2 - f \cdot y / z$$

Back-face cull
Skip polygons facing away. On for every opaque material.

Z-buffer
Keep the nearest depth at each pixel, order independent.

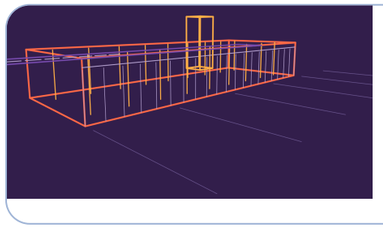
Glass
Alpha-blended with culling off to show what is behind.

A 3.6 m door at 10 m depth is about 336 px tall (1080p, 60° view)

6 Controls

WASD	move	E	talk to nearest
Mouse	orbit camera	R	respawn
Wheel	camera zoom	M	voices on/off
Shift	sprint	L	performance
Space	jump	Esc	exit

Run it: open the script in Blender's Scripting workspace (Blender 3.2+) and press **Alt+P**.



Perspective wireframe of the terminal

Drawn with the same projection maths the project is about: PROJECTION · BACK-FACE CULLING · Z-BUFFER

One Python script builds the airport and runs the game

Scene builders create the terminal, tower, runway, aircraft and cars. **Articulated people** use hip, knee, shoulder, elbow and head pivots. **Game layer** handles NPC behaviour, subtitles, voice lines and PA announcements.

Conclusion

Projection maps 3D coordinates to the screen with one divide by depth, and culling plus the depth buffer keep the nearest surface. A short mission makes it explorable.